

TESTING AND EVALUATION REPORT

AI-BASED FAKE PROFILE DETECTION

1. TESTING OBJECTIVES

Testing verifies correctness, reliability, security, model quality and usability.

2. UNIT TESTING

Test missing-value handling, ratios, categorical encoding, feature calculations, risk thresholds and API schemas.

3. INTEGRATION TESTING

Verify communication among frontend, API, model, explanation service and database.

4. SYSTEM TESTING

Scenario: analyst submits valid profile; system validates it, generates features, predicts probability, calculates risk, returns explanation and displays the result.

5. NEGATIVE TESTING

Invalid numeric values, missing mandatory fields, malformed JSON, oversized requests and unavailable model files should produce controlled errors.

6. SECURITY TESTING

Test authentication, authorization, rate limiting, injection attempts, secret exposure and excessive logging.

7. MODEL EVALUATION

Use a held-out test dataset. Calculate:

Accuracy = correct predictions / total predictions.

Precision = $TP / (TP + FP)$.

Recall = $TP / (TP + FN)$.

F1 = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$.

ROC-AUC can be used to evaluate ranking quality across classification thresholds.

8. CONFUSION MATRIX

True Positive: fake account correctly identified.

True Negative: genuine account correctly identified.

False Positive: genuine account incorrectly flagged.

False Negative: fake account missed.

9. MODEL COMPARISON

Evaluate Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and optionally XGBoost. Record Accuracy, Precision, Recall, F1, ROC-AUC and inference time.

10. RESULT REPORTING

Actual numerical results must come from the selected dataset. The report should include charts for class distribution, feature distributions, confusion matrix and model comparison.

11. ACCEPTANCE TESTING

The system passes when valid requests produce correct-format results, invalid requests are rejected, model evaluation is reproducible, and the dashboard presents understandable output.

12. LIMITATIONS

Dataset bias, label quality, concept drift, adversarial behavior and false positives can affect real-world performance.